Oxidation Numbers and Redox Reactions

Answer key with worked solutions

A. Understanding

1. Definitions.

> **Oxidation** is the loss of electrons; the oxidation number **increases**. **Reduction** is the gain of electrons; the oxidation number **decreases**.

B. Oxidation numbers

2. Oxidation numbers.

> (a) **−3** (b) **+5** (c) **+6** (d) **+6** (e) **+3** (f) **−1** (peroxide) (g) **+4** (h) **+5**. For example, (h): x + 3(−2) = −1, so x = +5.

C. Agents and half-equations

3. Chlorine and bromide.

> **Br^- is oxidized** (−1 → 0) and **Cl_2 is reduced** (0 → −1). **Oxidizing agent: Cl_2; reducing agent: Br^-.**

4. Half-equations.

> Oxidation: $Mg \longrightarrow Mg^{2+} + 2\,e^-$. Reduction: $2\,H^+ + 2\,e^- \longrightarrow H_2$.

5. Redox or not?

> (a) **Not redox**: no oxidation number changes. (b) **Redox** (disproportionation): O goes from −1 to −2 in H_2O and to 0 in O_2. (c) **Not redox**: a precipitation in which all oxidation numbers stay the same.

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.

D. Balancing

6. Copper and silver.

> Cu loses 2 e^-; each Ag^+ gains 1 e^-, so 2 Ag^+ are needed: $Cu + 2\,Ag^+ \longrightarrow Cu^{2+} + 2\,Ag$. Charge: +2 = +2. ✓

7. Dichromate and iron(II).

> Reduction: $Cr_2O_7{}^{2-} + 14\,H^+ + 6\,e^- \longrightarrow 2\,Cr^{3+} + 7\,H_2O$
>
> Oxidation (× 6): $6\,Fe^{2+} \longrightarrow 6\,Fe^{3+} + 6\,e^-$
>
> Overall: $Cr_2O_7{}^{2-} + 14\,H^+ + 6\,Fe^{2+} \longrightarrow 2\,Cr^{3+} + 7\,H_2O + 6\,Fe^{3+}$. Charge: −2 + 14 + 12 = +24 on the left; +6 + 18 = +24 on the right. ✓

8. Copper and nitrate.

> Oxidation (× 3): $3\,Cu \longrightarrow 3\,Cu^{2+} + 6\,e^-$
>
> Reduction (× 2): $2\,NO_3{}^- + 8\,H^+ + 6\,e^- \longrightarrow 2\,NO + 4\,H_2O$
>
> Overall: $3\,Cu + 2\,NO_3{}^- + 8\,H^+ \longrightarrow 3\,Cu^{2+} + 2\,NO + 4\,H_2O$. Charge: −2 + 8 = +6 = +6. ✓

E. Challenge

9. Permanganate.

> (a) Reduction (× 2): $2\,MnO_4{}^- + 16\,H^+ + 10\,e^- \longrightarrow 2\,Mn^{2+} + 8\,H_2O$; oxidation (× 5): $5\,C_2O_4{}^{2-} \longrightarrow 10\,CO_2 + 10\,e^-$
>
> Overall: $2\,MnO_4{}^- + 5\,C_2O_4{}^{2-} + 16\,H^+ \longrightarrow 2\,Mn^{2+} + 10\,CO_2 + 8\,H_2O$. Charge: −2 − 10 + 16 = +4 = +4. ✓
>
> (b) $n(MnO_4{}^-) = 0.0200\ \text{mol/L} \times 0.02000\ \text{L} = 4.00 \times 10^{-4}\ \text{mol}$
>
> $$n(Fe^{2+}) = 4.00 \times 10^{-4}\ \text{mol}\ MnO_4{}^- \times \frac{5\ \text{mol}\ Fe^{2+}}{1\ \text{mol}\ MnO_4{}^-} = 2.00 \times 10^{-3}\ \text{mol}$$
>
> $$[Fe^{2+}] = \frac{2.00 \times 10^{-3}\ \text{mol}}{0.02500\ \text{L}} = \mathbf{0.0800\ M}$$

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.

10. Basic solution.

Oxidation (in acid, × 2): $2\,Cr(OH)_3 + 2\,H_2O \longrightarrow 2\,{CrO_4}^{2-} + 10\,H^+ + 6\,e^-$

Reduction (in acid, × 3): $3\,ClO^- + 6\,H^+ + 6\,e^- \longrightarrow 3\,Cl^- + 3\,H_2O$

Add and cancel: $2\,Cr(OH)_3 + 3\,ClO^- \longrightarrow 2\,{CrO_4}^{2-} + 3\,Cl^- + 4\,H^+ + H_2O$

Add $4\,OH^-$ to both sides ($4\,H^+ + 4\,OH^- \longrightarrow 4\,H_2O$):

$2\,Cr(OH)_3 + 3\,ClO^- + 4\,OH^- \longrightarrow 2\,{CrO_4}^{2-} + 3\,Cl^- + 5\,H_2O$

Check: O: 6 + 3 + 4 = 13 and 8 + 5 = 13; H: 6 + 4 = 10 and 10; charge: −3 − 4 = −7 and −4 − 3 = −7. ✓

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.